

Edge Task Hub

Resource-aware Edge AI automation on Raspberry Pi.

- Feishu is the message channel.
- Raspberry Pi is the intelligence boundary.
- Local inference, local scheduling, local fallback, and local monitoring happen on the edge device.

What We Must Prove

This is an Edge AI course project, not a general chatbot project.

- The presentation focuses on experimental setup, results, discussion, and analysis.
- Every result is explained through edge constraints.
- The key claim is resource-aware local intelligence, not simply using an LLM.

Why Edge AI?

Cloud AI can ignore constraints that matter on a Raspberry Pi.

- Privacy: private document text should stay on the device.
- Memory and storage: model choice must fit real Pi limits.
- Latency policy: reminders must be punctual; summaries can wait.
- Reliability: the system still needs useful behavior when an LLM fails.

System Architecture

Feishu is the interface; the Raspberry Pi owns the local decision path.

- FastAPI exposes local APIs and Feishu callback endpoints.
- APScheduler controls reminder and news-summary timing.
- SQLite records tasks, runs, events, document metadata, and metrics.
- Ollama and Isolation Forest provide local intelligence.

User Workflows

The project is a useful edge system, not only a model benchmark.

- Daily previous-day news summary: 0 10 * * * local time.
- Built-in lunch and sleep reminders: 12:00 and 23:30.
- User-created reminders through Feishu text commands.
- Private document summaries from Feishu file events.

Experimental Setup

Experiments were run on the deployed Raspberry Pi, not only on a laptop.

- Runtime stack: FastAPI, APScheduler, SQLite, Ollama, and scikit-learn.
- LLM candidates: qwen3.5:2b, qwen3.5:0.8b, qwen2.5:0.5b, qwen3:1.7b, and qwen3:0.6b.
- Non-LLM intelligence: deterministic rules fallback and Isolation Forest.

Model Selection Journey

Changing models was the experiment: each model tested a different edge constraint.

- qwen3.5:2b: stronger expected quality, but too heavy and unstable for Pi deployment.
- qwen3.5:0.8b: quantized, but still slow and output-unstable on cold start.
- qwen2.5:0.5b-instruct: fast, but lower summary quality.
- Final design: qwen3:1.7b for waitable summaries and qwen3:0.6b for shorter prompts.

Model Results

The final model choice balances quality, speed, memory, and deployability.

Candidate	Constraint	Decision
qwen3.5:2b	Storage and deployment stability	Rejected before final inference
qwen3:1.7b	Quality-first local summarization	Selected
qwen3:0.6b	Low-latency lightweight prompts	Kept as fast model
qwen3.5:0.8b	Memory pressure and cold-start latency	Rejected
qwen2.5:0.5b-instruct	Quality loss from very small LLM	Rejected for final summaries

Why qwen3.5:0.8b Was Rejected

Quantization helps model size, but it did not solve this edge workload.

- Recorded cold-start runs were about 52 to 75 seconds.
- Generation speed was about 2 tokens/s.
- Visible output was empty or length-limited.
- Likely causes: cold loading, Pi CPU speed, memory pressure, and thinking-token budget.

Why Non-LLM Models Matter

Edge AI should use the smallest reliable intelligence for each local job.

- Rules fallback is deterministic when Ollama is unavailable.
- Isolation Forest is better than an LLM for CPU, memory, disk, temperature, and network metrics.
- APScheduler plus SQLite gives local timing control for reminders.
- This is hybrid Edge AI, not only LLM Edge AI.

Privacy and Automation Results

Private processing stays local while Feishu only delivers commands and results.

- Feishu text can create scheduled reminders.
- Feishu files can trigger private document summaries.
- Document text is extracted and summarized locally.
- Verified checks cover Feishu parsing, slow-summary acceptance, document extraction, and English-only materials.

Discussion and Analysis

The strongest project argument is the edge tradeoff, not only the generated text.

- Daily summaries can wait for qwen3:1.7b quality.
- Reminders need local timing reliability.
- Private files need local processing.
- Device health needs lightweight local ML.
- One best model does not exist for all edge tasks.

Limitations and Conclusion

Edge Task Hub is a real Edge AI system with honest boundaries.

- Feishu public callback configuration is still required.
- Local LLM cold starts remain slow.
- Scanned PDFs need OCR in future work.
- Final claim: useful AI can run under edge constraints by choosing the right local model for each job.